

OE-EC704A — Web Technology

Semester VII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Web Development Basics (HTML & CSS)

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Contents

1	Introduction to Web Development Basics	2
1.1	HTML vs. XML	2
2	HTML Basics and Document Structure	3
2.1	Standard Skeleton of an HTML5 Document	3
2.2	Block-level vs. Inline Elements	3
2.3	Essential HTML Components	4
3	CSS (Cascading Style Sheets)	5
3.1	Types of CSS Integration	5
3.2	CSS Selectors	5
3.3	The CSS Box Model	5

1. Introduction to Web Development Basics

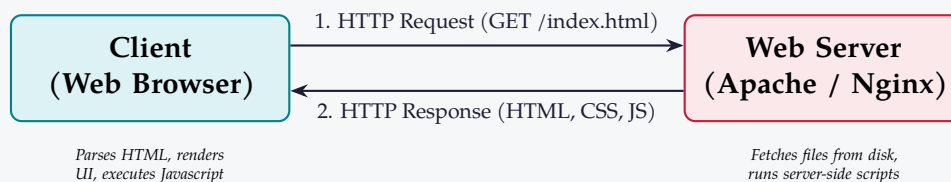
Modern web development relies on a client-server architecture. Web pages are hosted on web servers and requested by clients (web browsers) via HTTP/HTTPS protocols.

Client-Server Web Architecture

The **Client-Server Model** is a distributed application structure that partitions tasks between resource providers (**servers**) and service requesters (**clients**). In web technology:

- **Client (Web Browser):** Sends HTTP requests, parses HTML/CSS/JS, and renders the user interface.
- **Server (Web Server):** Listens for incoming requests on ports 80 (HTTP) or 443 (HTTPS), processes logic, and returns the requested resources.

Web Architecture: Client-Server Flow



1.1 HTML vs. XML

HTML and XML are both markup languages based on SGML, but they serve fundamentally different purposes:

Feature	HTML (HyperText Markup Language)	XML (eXtensible Markup Language)
Primary Purpose	Displaying and formatting data to users.	Transporting, storing, and structuring data.
Tags	Predefined tags (e.g., <html>, <p>).	Custom/user-defined tags based on data.
Syntax Strictness	Forgiving (errors don't crash rendering).	Extremely strict (well-formed check required).
Case Sensitivity	Case-insensitive (<DIV> and <div>).	Strictly case-sensitive.
Closing Tags	Optional for some tags (e.g., ,).	Mandatory for all tags.

2. HTML Basics and Document Structure

HTML — HyperText Markup Language

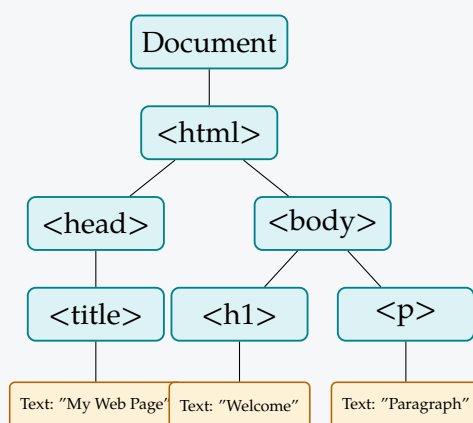
HTML is the standard markup language used to create the structure of web pages. It uses a set of hierarchical elements represented by tags to organize headings, paragraphs, tables, images, and other media elements.

2.1 Standard Skeleton of an HTML5 Document

Every HTML document must follow a standard structural hierarchy:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <title>My Web Page</title>
  </head>
  <body>
    <h1>Welcome to Web Technology</h1>
    <p>This is a paragraph.</p>
  </body>
</html>
```

HTML Document Object Model (DOM) Tree



2.2 Block-level vs. Inline Elements

HTML elements are categorized based on their default layout behaviors:

Feature	Block-level Elements	Inline Elements
Layout Behavior	Starts on a new line and takes up full width.	Stays on the same line and occupies only needed width.
Margins & Padding	Fully respected on all four sides.	Top and bottom margins are ignored.
Containment	Can contain block or inline elements.	Can only contain inline elements / text.
Examples	<code><div></code> , <code><p></code> , <code><h1></code> – <code><h6></code> , <code></code> , <code></code> , <code><table></code>	<code></code> , <code><a></code> , <code></code> , <code></code> , <code></code> , <code><input></code>

2.3 Essential HTML Components

1. **Lists:** HTML supports three list types:

- **Unordered List (`<ul>`):** Bulleted list items using the `<li>` tag.
- **Ordered List (`<ol>`):** Numbered list items with customizable types (numeric, alphabetic, roman).
- **Description List (`<dl>`):** Pairs of terms (`<dt>`) and descriptions (`<dd>`).

2. **Links (Anchors `<a>`):** Used for navigating between pages or sections:

- **Absolute Link:** Links to a fully qualified URL (e.g., `href="https://google.com"`).
- **Relative Link:** Links to a local resource relative to the file path (e.g., `href="about.html"`).
- **Email Link:** Triggers the client's email client using the mailto protocol (e.g., `href="mailto:test@test.com"`).
- **Anchor Targets:** Uses `target="_blank"` to open in a new tab or `target="_self"` for the same frame.

3. **Tables (`<table>`):** Formats data in rows and columns:

- **Rowspan:** Extends a cell vertically across multiple rows (e.g., `<td rowspan="2">`).
- **Colspan:** Extends a cell horizontally across multiple columns (e.g., `<td colspan="3">`).

4. **Forms (`<form>`):** Captures and submits user inputs to a server. Key attributes:

- **action:** The URL of the server-side endpoint that handles the submitted form data.
- **method:** The HTTP method used to send data (GET or POST).
- **Input Types:** Text, password, radio buttons, checkboxes, email, submit, file, number.

5. **Frames (`<frameset>`, `<frame>`, `<iframe>`):** Used to split the browser window into multiple independent document spaces:

- **Traditional Frameset (`<frameset>`):** Replaces the `<body>` tag in HTML4 to partition the screen layout into rows or columns (e.g., `<frameset cols="25%, 75%">`). Each partition is defined using a `<frame src="...">` tag. *Note: Deprecated in HTML5 due to poor accessibility, bookmarking difficulties, and bad SEO indexing.*

- **Inline Frames (<iframe>):** Used inside a standard <body> tag to embed another HTML page within the current page (e.g., <iframe src="inner.html" width="300" height="200"></iframe>). Widely used for embedding maps, videos, and third-party widgets.

3. CSS (Cascading Style Sheets)

CSS — Cascading Style Sheets

CSS is a style sheet language used to describe the presentation, design, formatting, and layout of a document written in HTML. It separates structure from style, allowing consistent look-and-feel across multiple web pages.

3.1 Types of CSS Integration

CSS Type	Syntax Example	Advantages	Disadvantages
Inline CSS	<code><p style="color:red;"></code>	High priority; quick overrides.	Messy; violates separation of concerns.
Internal CSS	<code><style> p { color: red; } </style></code>	Contained inside single page; no extra HTTP request.	Reusability limit across multiple pages.
External CSS	<code><link rel="stylesheet" href="style.css"></code>	High reuse; caching support; clean separation.	Requires extra network HTTP request.

3.2 CSS Selectors

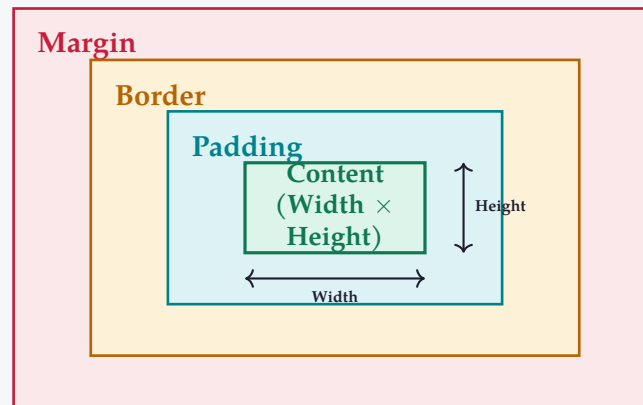
CSS selectors target specific HTML elements for styling:

- **Element Selector:** Targets all elements of a type (e.g., `p { color: red; }`).
- **Class Selector:** Targets elements with a matching class name, preceded by a dot (e.g., `.container { width: 80%; }`).
- **ID Selector:** Targets a single element with a matching ID, preceded by a hash (e.g., `#header { height: 60px; }`). ID must be unique within the page.
- **Universal Selector:** Targets every element in the DOM using asterisk (e.g., `* { margin: 0; }`).

3.3 The CSS Box Model

Every element in HTML is rendered as a rectangular box. The CSS Box Model defines how the size and spacing of this box are calculated.

CSS Box Model Layout



The dimensions of an element are computed based on the `box-sizing` property:

- **content-box (Default):** Total Width = width + left/right padding + left/right border + left/right margin.
- **border-box:** Total Width = width + left/right margin (padding and border are absorbed inside the specified width).

Quick Revision — Module 1

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Core HTML Tags & CSS Syntax

HTML Form / Tag	Syntax and Key Attributes
Hyperlink Anchor	<code>Link Text</code>
Embedded Image	<code></code>
HTML Table Structure	<code><table> <tr> <td colspan="2">Data</td> </tr> </table></code>
HTML Form Tag	<code><form action="/login" method="POST"> inputs... </form></code>
CSS Selector Type	Syntax and Example Usage
CSS Syntax Rule	<code>selector { property: value; }</code>
Class Styling	<code>.button { background-color: blue; padding: 10px; }</code>
ID Styling	<code>#main-nav { display: flex; justify-content: space-between; }</code>
Universal Reset	<code>* { box-sizing: border-box; margin: 0; padding: 0; }</code>

One-Line Definitions

- **HTML:** Structured markup language defining content hierarchy of a web document.
- **CSS:** Style sheet language used for presenting and laying out HTML elements.
- **DOM (Document Object Model):** An API representing an HTML document as a logical tree structure.
- **Tag:** A code marker representing the start or end of an HTML element (e.g., `<p>`).
- **Element:** A complete component consisting of start tag, content, and end tag.
- **Attribute:** Name-value pairs inside HTML tags that modify their behavior or supply metadata.
- **Viewport:** The visible area of a web page on a client device screen.
- **URL (Uniform Resource Locator):** The global reference address of a web resource on the Internet.

Frequently Asked / Viva Questions

Q1. What is the difference between **GET** and **POST** methods in forms?

GET appends form data directly to the URL as query parameters, rendering it visible and bookmarkable. It is intended for idempotent requests. **POST** sends data inside the HTTP message body, making it secure for sensitive parameters like passwords.

Q2. Why is **DOCTYPE html** declared at the beginning of an HTML file?

Frequently Asked / Viva Questions (contd.)

It is a standard preamble instructing the client browser to render the document in standards-compliant mode rather than quirks mode.

Q3. What is the difference between `span` and `div` tags?

`div` is a block-level container element that starts on a new line. `span` is an inline element used to style specific words or content inline without inserting page breaks.

Q4. How does CSS specificity calculate which style to apply?

Specificity is calculated as a numeric weight: inline styles hold the highest priority (1000), followed by ID selectors (100), class/attribute selectors (10), and element selectors (1).

Q5. What is the difference between relative and absolute positioning in CSS?

Relative positioning offsets an element relative to its normal position in the document flow. Absolute positioning moves the element relative to its closest parent with non-static positioning.

Q6. What does the `box-sizing: border-box` property do?

It alters the CSS Box Model calculation so that borders and paddings are included in the element's declared width and height, preventing layout breakage.

Q7. What are `colspan` and `rowspan` attributes in HTML tables?

`colspan` merges cells horizontally across columns. `rowspan` merges cells vertically across rows.

Q8. What is semantic HTML and why is it important?

Semantic HTML uses tags that explain their purpose clearly (e.g., `<header>`, `<nav>`, `<article>`, `<footer>`) instead of generic `<div>` blocks. It enhances accessibility and SEO search indexing.

Q9. How do you link CSS files to an HTML document?

Using the HTML `<link>` element inside the `<head>` block: `<link rel="stylesheet" href="style.css">`.

Q10. What are CSS pseudoclasses and how are they used?

Pseudoclasses define special states of elements (e.g., `:hover` for mouse hover, `:active` for clicks, `:focus` for keyboard selection).

Form Submission: GET vs. POST comparison

Metric	GET Method	POST Method
Data Location	Appended to URL (Query parameters).	Inside HTTP Request Body.
Security	Poor (exposed in browser history/logs).	Higher (hidden from URL path).
Payload Limits	Limited (typically 2048 characters).	No architectural payload size limit.
Caching	Allowed/Cached by proxy servers.	Never cached.
Idempotency	Idempotent (submitting multiple times is safe).	Non-idempotent (resubmitting duplicates actions).